

NB35: PAH Panel

Literatur-Destekli Feature Engineering

Ablasyon Raporu

Tarih: 2026-06-24 10:55

Baseline: NB21 P4_COMBINED_BalBag

MCC=0.529, Boot-F1=0.582

TN=46, FP=16, FN=45, TP=262

Training: COMBINED(3430) + BalancedBagging(20xLGBM)

Panel: PAH (372 rows -> 369 after dedup, 310 patho / 62 benign)

1. Motivasyon ve Arka Plan

PAH paneli projenin en zorlu parçasıdır. Temel nedenler: n=62 benign (azınlık-ornegi tavani), zayıf EK korelasyonları (max $|r|=0.284$), MASTER ile gen-profil farkı. Feature engineering, model degistirmeden sinyal/gurultu oranini artiran yontemdir. NB34 CFTR'de benzer yaklasim test edilmisti. PAH'ta ayni literatur-destekli FE kullanilir.

2. Feature Gruplari Detayi (21 feature, 5 grup)

Grup A (2): Frekans - populasyon frekans bilgisi, ACMG BA1/BS1 kriterleri

Grup B (5): EK Skor Birlesimleri - meta-prediktor kombinasyonlari

Grup D (5): AA Fizikokimyasal Delta - disorder/accessibility yeni eklendi

Grup E (5): AA Substitusyon Skorlari - Grantham, BLOSUM62, proline

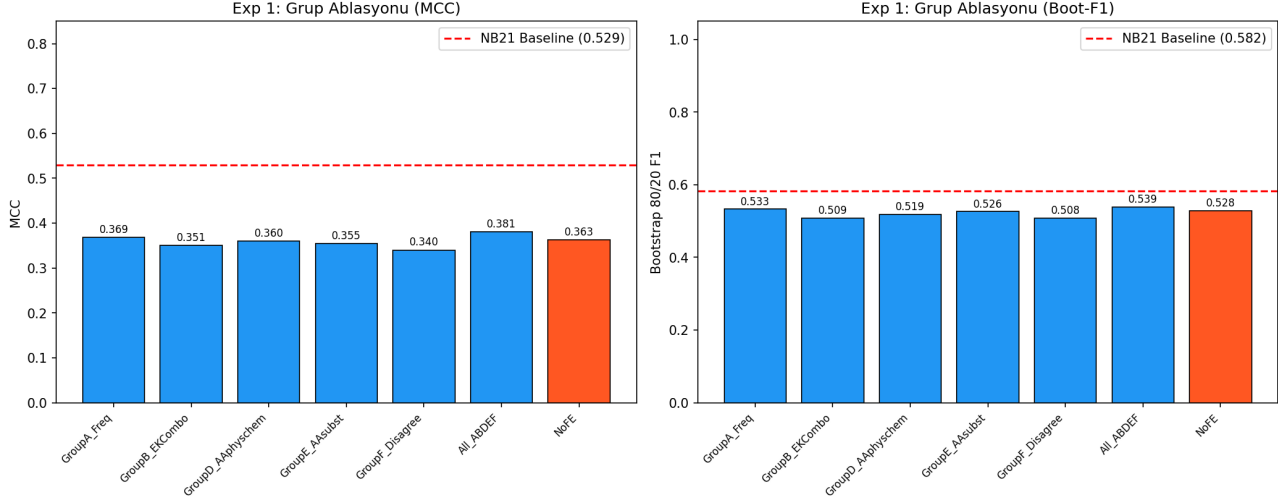
Grup F (4): Prediktor Uyumsuzlugu - PAH-spesifik, yeni grup

Kaynak: Grantham 1974, MutPred2, PON-P3, BMPR2 structural studies

3. Exp 1 - Grup Ablasyonu

Deney	MCC	Boot-F1	Prec	FP	FN
E1a_GroupA_Freq	0.3686	0.5329	0.942	13	95
E1b_GroupB_EKCombo	0.3508	0.5086	0.938	14	97
E1c_GroupD_AAphyschem	0.3596	0.5187	0.938	14	94
E1d_GroupE_AASubst	0.3546	0.5261	0.944	12	104
E1e_GroupF_Disagree	0.3399	0.5080	0.940	13	105
E1f_All_ABDEF	0.3807	0.5388	0.950	11	99
E1g_NoFE	0.3627	0.5284	0.942	13	97

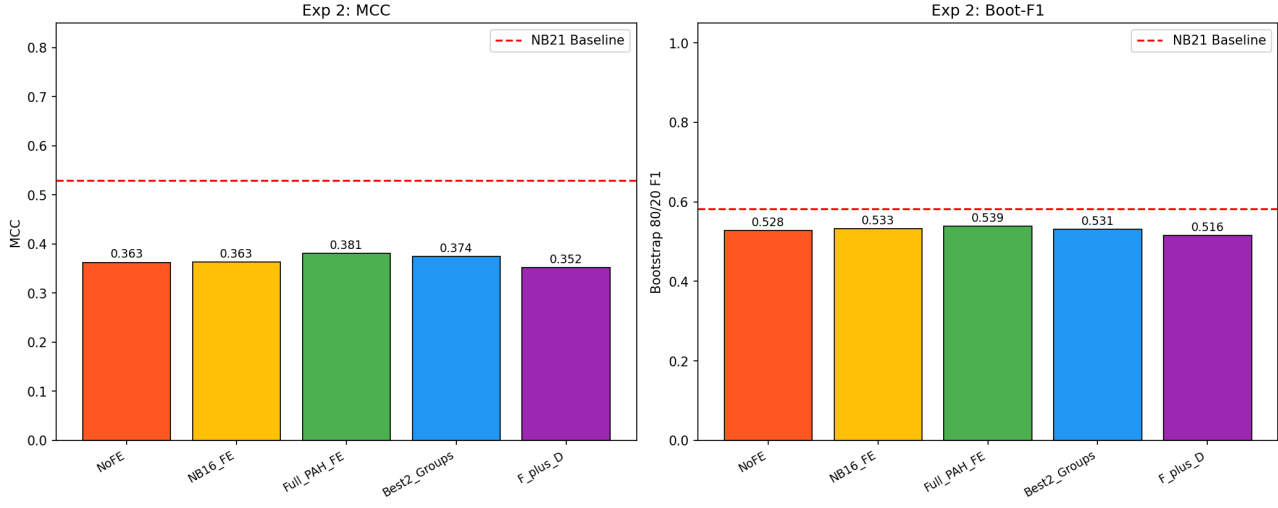
NB35 - PAH Feature Engineering Ablation



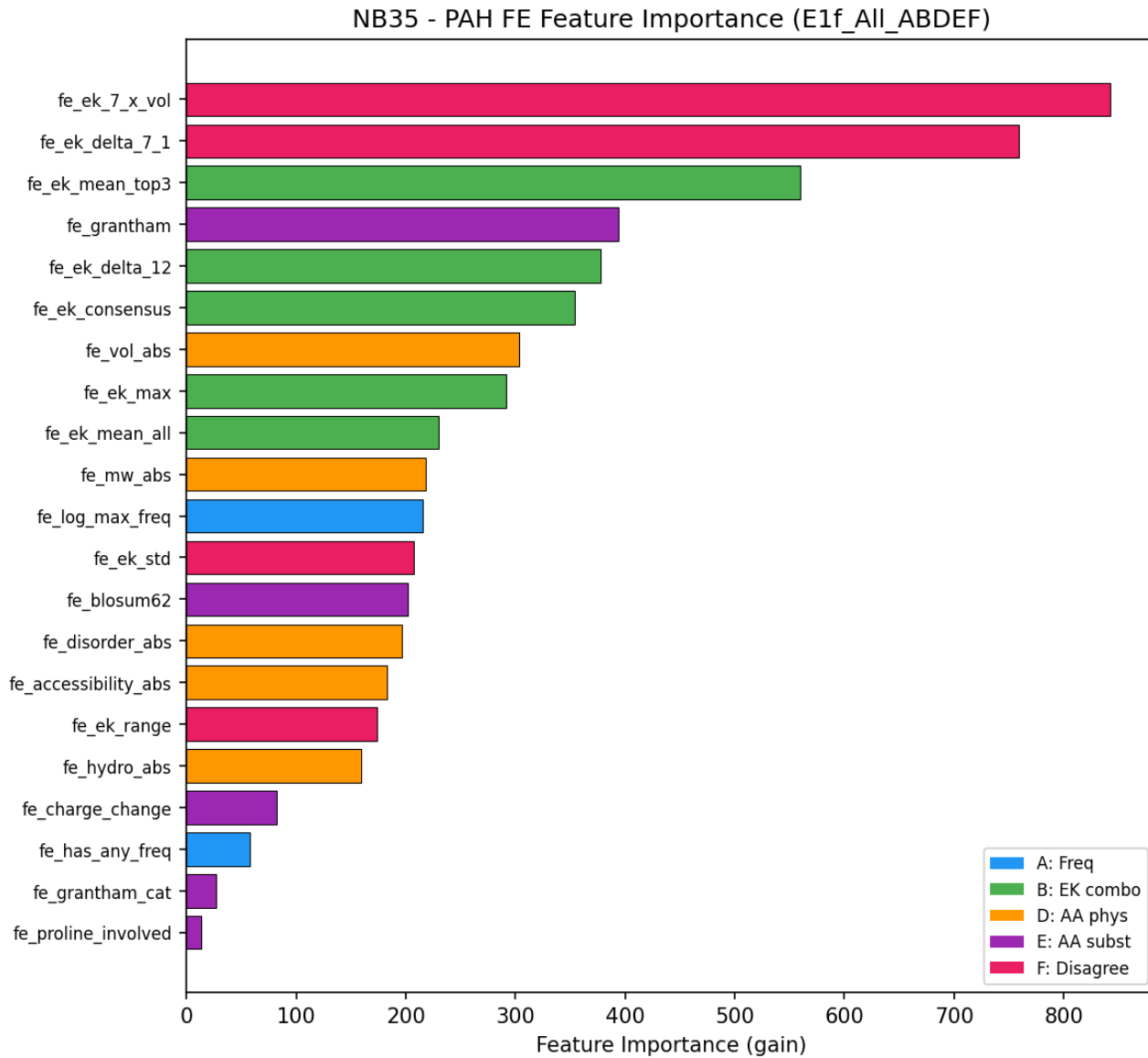
4. Exp 2 - FE Versiyon Karsilastirmasi

Deney	MCC	Boot-F1	n_feat
E2a_NoFE	0.3627	0.5284	428
E2b_NB16_FE	0.3631	0.5328	430
E2c_Full_PAH_FE	0.3807	0.5388	449
E2d_Best2_Groups	0.3745	0.5312	435
E2e_F_plus_D	0.3518	0.5164	437

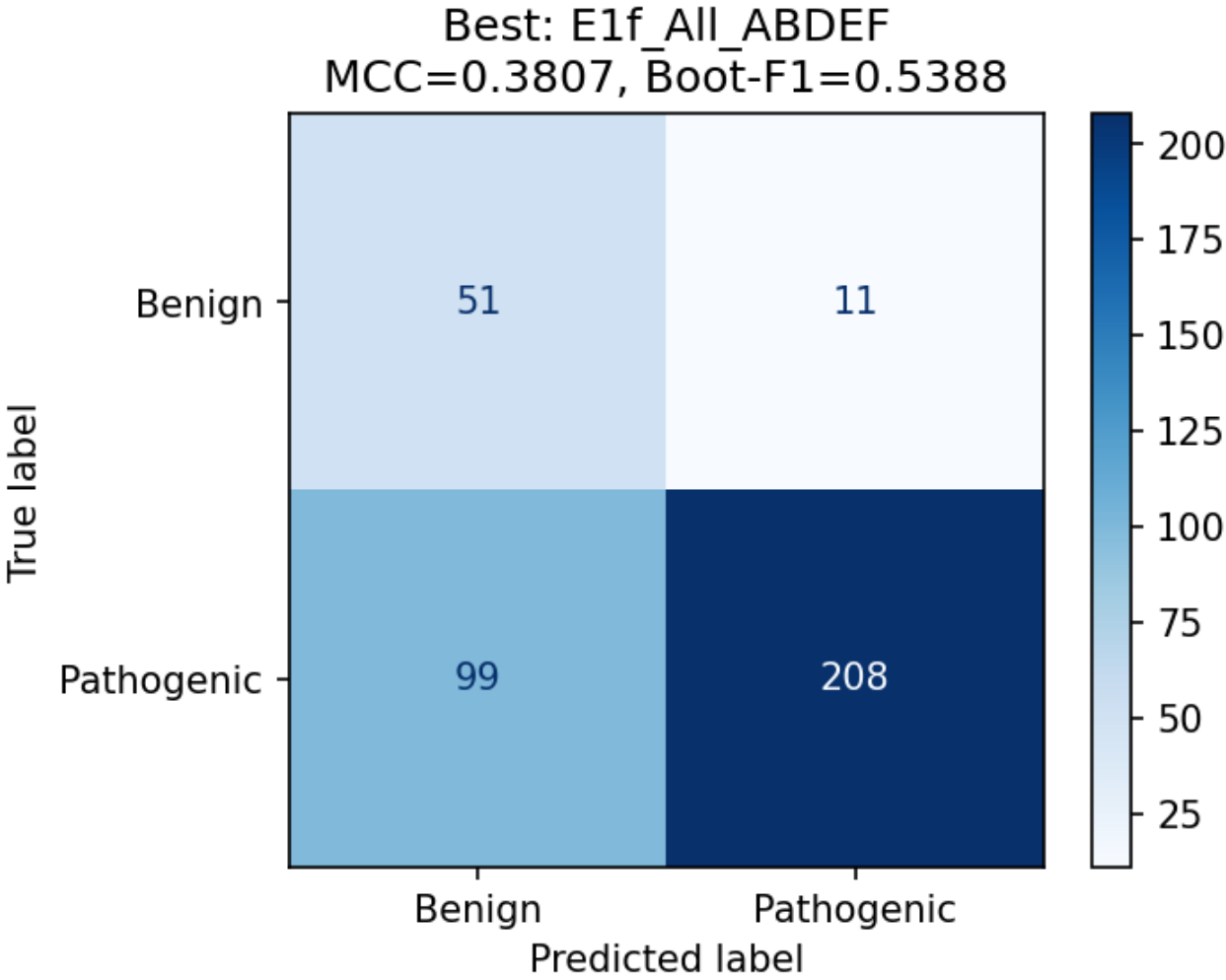
NB35 - PAH FE Version Comparison



5. Feature Importance Analizi

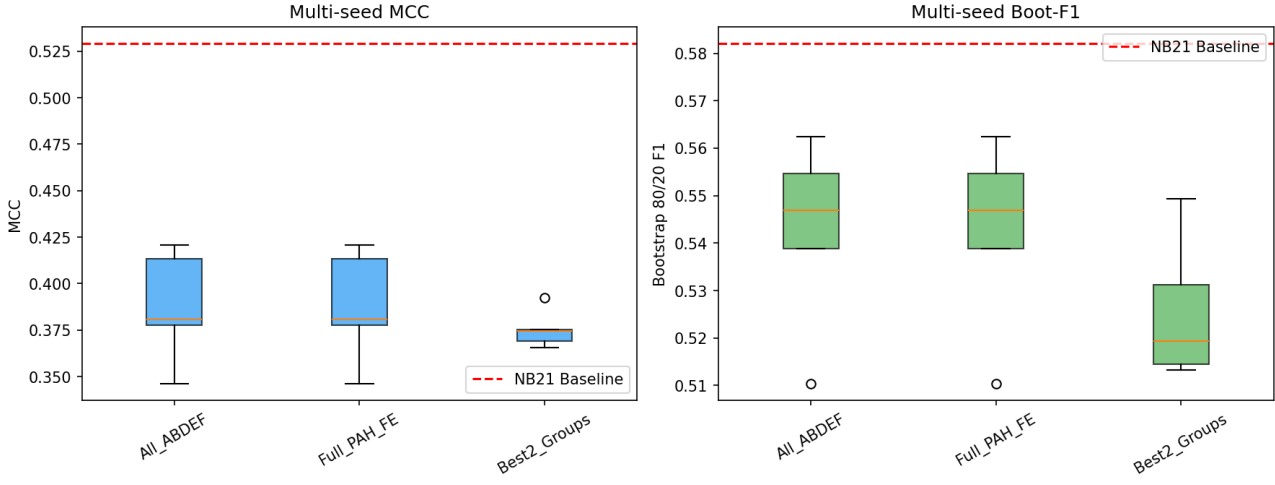


6. Confusion Matrix



7. Multi-seed Dogrulama

NB35 - PAH Multi-seed Validation



8. Sonuc ve Karar

En iyi model: E1f_All_ABDEF

MCC: 0.3807 vs 0.529 (NB21), Delta=-0.1483

Boot-F1: 0.5388 vs 0.582 (NB21)

Karar: FE deger katmadi

PAH'in temel zorluklari FE ile tam cozulmeyebilir: n=62 benign tavani ve zayif EK korelasyonlari.